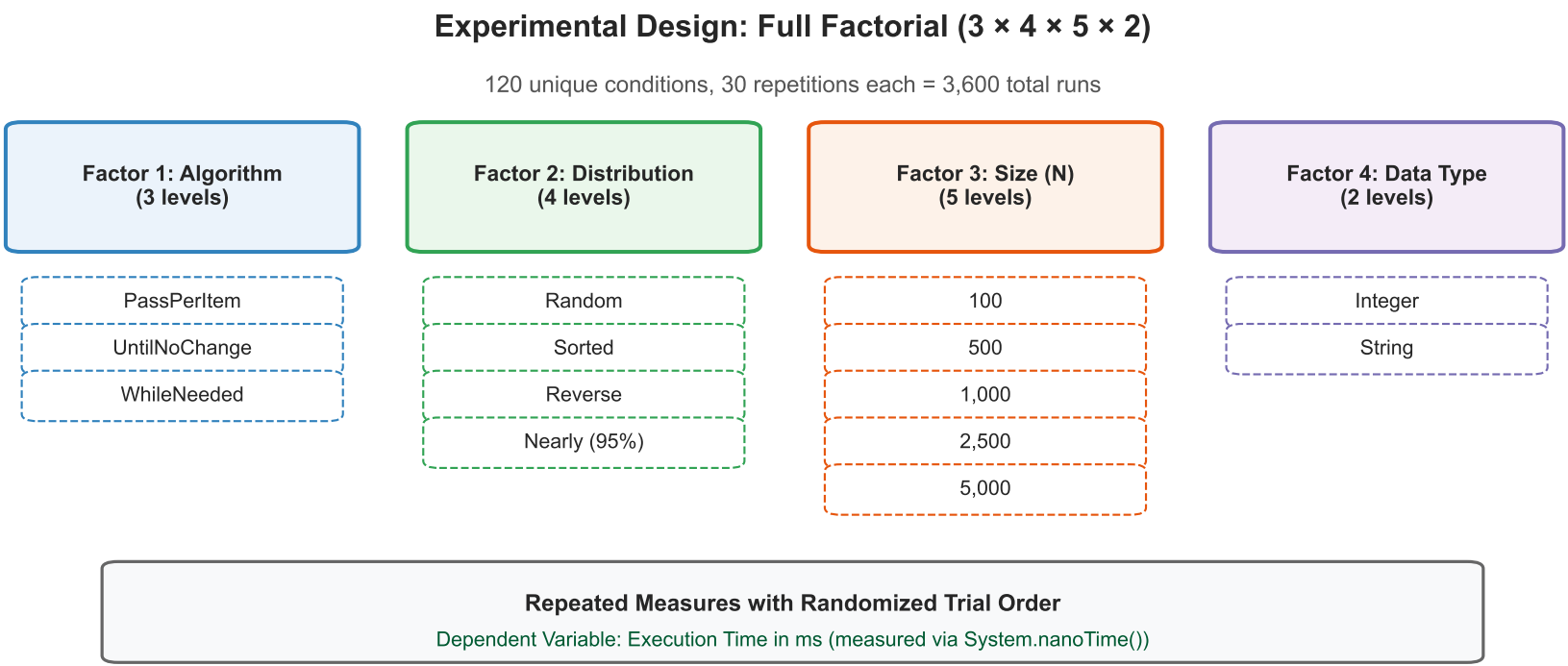
Experimental Design: Full Factorial (3 × 4 × 5 × 2)
120 unique conditions, 30 repetitions each = 3,600 total runs
Factor 1: Algorithm (3 levels)
PassPerItem
UntilNoChange
WhileNeeded
Factor 2: Distribution (4 levels)
Random
Sorted
Reverse
Nearly (95%)
Factor 3: Size (N) (5 levels)
100
500
1,000
2,500
5,000
Factor 4: Data Type (2 levels)
Integer
String
Repeated Measures with Randomized Trial Order
Dependent Variable: Execution Time in ms (measured via System.nanoTime())